Chapter 6. Estimation

Method of moments, maximum likelihood, multiparameter MLE, asymptotics of the MLE, least squares

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Setting

$X_1, \ldots, X_n$ is a random sample from $f(x;\theta)$, $\theta \in \Omega$. A statistic $\hat{\eta} = \hat{\eta}(X_1, \ldots, X_n)$ used to guess $\eta = \eta(\theta)$ is an **estimator**; its value at the data is an **estimate**. P_θ, E_θ, Var_θ are computed under parameter θ.

Definition (Consistency)

$\hat{\eta}_n$ is **consistent** for $\eta(\theta)$ if $\hat{\eta}_n \xrightarrow{p} \eta(\theta)$ under P_θ for every $\theta \in \Omega$, i.e. $\lim_n P_\theta(|\hat{\eta}_n - \eta(\theta)| \geq \epsilon) = 0$ for all $\epsilon > 0$ and all θ.

6.1 The method of moments

The population moment $m_r = EX_1^r$ is estimated by $\hat{m}_r = \frac{1}{n}\sum X_i^r$. If $\eta = g(m_1, \ldots, m_k)$, the **method-of-moments estimator** (MME) is

$$\hat{\eta}^{\mathrm{MME}} = g(\hat{m}_1, \ldots, \hat{m}_k).$$

Multivariate populations use joint moments $m_{r,s} = E(X_1^r Y_1^s)$ and $\hat{m}_{r,s} = \frac{1}{n}\sum X_i^r Y_i^s$.

Example 6.1

(a) $\sigma^2 = m_2 - m_1^2$, so

$$\hat{\sigma}^{2\,\mathrm{MME}} = \hat{m}_2 - \hat{m}_1^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2, \qquad \hat{\sigma}^{\mathrm{MME}} = \sqrt{\frac{1}{n}\sum (X_i - \bar{X})^2}.$$

(b) The MME is not unique: for Poisson(λ), $\lambda = m_1 = m_2 - m_1^2$ gives $\bar{X}$ and $\frac{1}{n}\sum(X_i - \bar{X})^2$. Use the lowest moments possible.

(c) For a bivariate population, $\rho = \dfrac{m_{1,1} - m_{1,0}m_{0,1}}{\sqrt{m_{2,0} - m_{1,0}^2}\sqrt{m_{0,2} - m_{0,1}^2}}$, so

$$\hat{\rho}^{\mathrm{MME}} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2}\sqrt{\sum (Y_i - \bar{Y})^2}},$$

the sample correlation coefficient.

(d) Gamma(α, β): $m_1 = \alpha\beta$, $m_2 - m_1^2 = \alpha\beta^2$ give $\hat{\beta} = \hat{\sigma}^2/\bar{X}$, $\hat{\alpha} = \bar{X}^2/\hat{\sigma}^2$, $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.

Theorem 6.1 (Consistency of MMEs)

If $E|X_1|^k < \infty$ and g is continuous at $(m_1, \ldots, m_k)$, then $\hat{\eta}_n^{\mathrm{MME}} = g(\hat{m}_1, \ldots, \hat{m}_k) \xrightarrow{p} g(m_1, \ldots, m_k) = \eta$.

Proof. By the multivariate LLN $(\hat{m}_1, \ldots, \hat{m}_k)' \xrightarrow{p} (m_1, \ldots, m_k)'$ (Example 5.5 in Volume I), and continuous maps preserve convergence in probability (Theorem 5.7 in Volume I). ■

So $\hat{\sigma}^{2\,\mathrm{MME}}$, $S^2 = \frac{n}{n-1}\hat{\sigma}^{2\,\mathrm{MME}}$ and the sample correlation are consistent. MMEs are also asymptotically normal:

Theorem 6.2 (Limiting distribution of MMEs)

If $EX_1^{2k} < \infty$ and g has continuous first partial derivatives at $\mathbf{m} = (m_1, \dots, m_k)'$, then

$$\sqrt{n}(\hat{\eta}_n^{\text{MME}} - \eta) \xrightarrow{d} N(0,\ \dot{g}(\mathbf{m})'\Sigma\,\dot{g}(\mathbf{m})), \qquad \Sigma = (m_{r+s} - m_r m_s)_{r,s=1}^{k}.$$

Proof. $(X_i, X_i^2, \dots, X_i^k)'$ are i.i.d. with mean $\mathbf{m}$ and covariance Σ ($\text{Cov}(X_1^r, X_1^s) = m_{r+s} - m_r m_s$); apply the multivariate CLT and the delta method (Theorem 5.12 in Volume I). ■

Example 6.2

$g(m_1, m_2) = m_2 - m_1^2$, $\dot{g} = (-2m_1, 1)'$, and $\dot{g}'\Sigma\dot{g} = \mu_4 - \sigma^4$, $\mu_4 = E(X_1 - \mu)^4$ (cf. Example 5.10 in Volume I).

6.2 Maximum likelihood

The sample falls in a small box around $\mathbf{x}$ with probability about $\prod_i f(x_i;\theta)\,|\Delta\mathbf{x}|$, so comparing $\prod f(x_i;\theta)$ across θ ranks parameter values by plausibility: regard the joint pdf as a function of θ.

Definition (Likelihood and MLE)

(a) The **likelihood function** is $L(\theta) = L(\theta;\mathbf{x}) = \prod_{i=1}^n f(x_i;\theta)$, $\theta \in \Omega$, and $l(\theta) = \log L(\theta) = \sum \log f(x_i;\theta)$ is the **log-likelihood**.

(b) A value $\hat{\theta}(\mathbf{x}) \in \Omega$ with $L(\hat{\theta}(\mathbf{x});\mathbf{x}) = \max_{\theta\in\Omega} L(\theta;\mathbf{x})$ is a **maximum likelihood estimate**; the statistic $\hat{\theta}^{\text{MLE}} = \hat{\theta}(\mathbf{X})$ is the **maximum likelihood estimator** (MLE).

$\dot{l}(\theta) = 0$ is the **likelihood equation**; the MLE is usually among its roots.

Example 6.3 (Poisson)

Poisson(λ), $\lambda \geq 0$ ($\lambda = 0$: point mass at 0):

$$l(\lambda) = -n\lambda + n\bar{x}\log\lambda - \log(x_1!\cdots x_n!), \qquad \dot{l}(\lambda) = -n + \frac{n\bar{x}}{\lambda}, \qquad \ddot{l}(\lambda) = -\frac{n\bar{x}}{\lambda^2}.$$

If $\bar{x} > 0$, l increases up to $\bar{x}$ and then decreases; if $\bar{x} = 0$, $l = -n\lambda$ decreases. By continuity of L on $[0,\infty)$, $\hat{\lambda}^{\text{MLE}} = \bar{X}$ in both cases.

Theorem 6.3

Let l be twice differentiable with continuous second derivative on an open interval Ω_o. If $\ddot{l}(\theta) < 0$ for all $\theta \in \Omega_o$ and $\dot{l}(\hat{\theta}) = 0$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of l on Ω_o.

Proof. By Taylor, with θ^* between θ and $\hat{\theta}$,

$$l(\theta) = l(\hat{\theta}) + \dot{l}(\hat{\theta})(\theta - \hat{\theta}) + \tfrac{1}{2}\ddot{l}(\theta^*)(\theta - \hat{\theta})^2 = l(\hat{\theta}) + \tfrac{1}{2}\ddot{l}(\theta^*)(\theta - \hat{\theta})^2 < l(\hat{\theta})$$

for $\theta \neq \hat{\theta}$. ■

Example 6.4 (Bernoulli)

Bernoulli(p), $0 \le p \le 1$, $\hat{p} = \bar{x}$:

$$l(p) = n\hat{p}\log p + n(1-\hat{p})\log(1-p), \qquad \dot{l}(p) = \frac{n\hat{p}}{p} - \frac{n(1-\hat{p})}{1-p}, \qquad \ddot{l}(p) = -\frac{n\hat{p}}{p^2} - \frac{n(1-\hat{p})}{(1-p)^2}.$$

If $0 < \hat{p} < 1$, $\ddot{l} < 0$ and $\dot{l}(\hat{p}) = 0$ (Theorem 6.3); if $\hat{p} = 0$, $L = (1-p)^n$ peaks at 0; if $\hat{p} = 1$, $L = p^n$ at 1. So $\hat{p}^{\mathrm{MLE}} = \bar{X}$.

$\ddot{l} < 0$ means l is **strictly concave**. When the likelihood equation has no explicit solution:

Theorem 6.4

Let l be twice continuously differentiable on an open interval Ω_o, with $\ddot{l}(\theta) < 0$ on Ω_o and $l(\theta) \to -\infty$ as θ tends to either end of Ω_o. Then the likelihood equation $\dot{l}(\theta) = 0$ has exactly one root $\hat{\theta}$ in Ω_o, and $\hat{\theta}$ is the unique maximizer of l.

Proof. Take $\Omega_o = \mathbb{R}$ (others similar). As $l \to -\infty$ at both ends, l attains its maximum; at two points $\hat{\theta}_1 < \hat{\theta}_2$ strict concavity would give

$$l\left(\tfrac{1}{2}\hat{\theta}_1 + \tfrac{1}{2}\hat{\theta}_2\right) > \tfrac{1}{2}l(\hat{\theta}_1) + \tfrac{1}{2}l(\hat{\theta}_2) = \max l,$$

a contradiction. At the maximum $\dot{l}(\hat{\theta}) = 0$, and $\dot{l}$ is strictly decreasing. ■

Without a closed form, use **Newton–Raphson**: from $0 = \dot{l}(\hat{\theta}) \approx \dot{l}(\theta^{(r)}) + \ddot{l}(\theta^{(r)})(\hat{\theta} - \theta^{(r)})$,

$$\hat{\theta}^{(r+1)} = \hat{\theta}^{(r)} + \left[-\ddot{l}(\hat{\theta}^{(r)})\right]^{-1}\dot{l}(\hat{\theta}^{(r)}), \qquad r = 0, 1, \ldots,$$

started e.g. at the MME.

Example 6.5 (Logistic location)

$f(x;\theta) = e^{-(x-\theta)}/(1+e^{-(x-\theta)})^2$, $\theta \in \mathbb{R}$:

$$l(\theta) = -n\bar{x} + n\theta - 2\sum_{i=1}^{n}\log\left(1 + e^{-(x_i-\theta)}\right), \qquad \dot{l}(\theta) = n - 2\sum_i \frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}},$$

$$\ddot{l}(\theta) = -2\sum_i \frac{e^{-(x_i-\theta)}}{(1+e^{-(x_i-\theta)})^2} < 0.$$

As $\theta \to -\infty$, $l \approx n\theta - n\bar{x} \to -\infty$; as $\theta \to \infty$, $l \approx -n\theta + n\bar{x} \to -\infty$. By Theorem 6.4 the MLE is the unique root; start Newton–Raphson at $\bar{x}$ (the logistic mean is θ).

Theorem 6.5 (Invariance)

If $\hat{\theta}^{\mathrm{MLE}}$ exists and $\eta = g(\theta)$ is a one-to-one function on Ω, then $\hat{\eta}^{\mathrm{MLE}} = g(\hat{\theta}^{\mathrm{MLE}})$.

Proof. The density with parameter η is $f(\cdot; g^{-1}(\eta))$, so $L^*(\eta) = L(g^{-1}(\eta))$ and

$$\max_{\eta \in g(\Omega)} L^*(\eta) = \max_{\theta \in \Omega} L(\theta) = L(\hat{\theta}^{\mathrm{MLE}}) = L^*(g(\hat{\theta}^{\mathrm{MLE}})).$$

■

Example 6.6 (Exponential)

$f(x;\theta)=\theta^{-1}e^{-x/\theta}$: $l(\theta)=-n\log\theta-n\bar{x}/\theta$ has $\ddot{l}(\theta)=n/\theta^2-2n\bar{x}/\theta^3$, not always negative. With $\lambda=1/\theta$, $l=n\log\lambda-n\bar{x}\lambda$ is strictly concave and $\to-\infty$ at $0,\infty$ ($\bar{x}>0$): $\hat{\lambda}=1/\bar{X}$, so $\hat{\theta}^{\mathrm{MLE}}=\bar{X}$ by invariance.

A good parametrization can make l concave; exponential families have a canonical one.

Theorem 6.6 (MLE in a one-parameter exponential family)

Let

$$f(x;\eta)=\exp\{\eta T(x)-A(\eta)+S(x)\},\qquad x\in\mathcal{X},\ \eta\in N,$$

where (i) the support $\mathcal{X}$ does not depend on η;
(ii) N is an open interval;
(iii) $\mathrm{Var}_\eta\, T(X_1)>0$. If the likelihood equation

$$\dot{A}(\eta)=\frac{1}{n}\sum_{i=1}^{n}T(x_i),\qquad \text{i.e.}\quad E_\eta T(X_1)=\frac{1}{n}\sum_{i=1}^{n}T(x_i),$$

has a root $\hat{\eta}$ in N, then $\hat{\eta}$ is the MLE of η.

Proof. The cgf of $T(X_1)$: for small $|s|$ ($s+\eta\in N$ by (ii)), in the continuous case

$$E_\eta e^{sT(X_1)}=\int e^{(s+\eta)T(x)-A(\eta)+S(x)}dx=e^{A(s+\eta)-A(\eta)}\int f(x;s+\eta)\,dx=e^{A(s+\eta)-A(\eta)},$$

using (i). So $E_\eta T(X_1)=\dot{A}(\eta)$ and $\mathrm{Var}_\eta\, T(X_1)=\ddot{A}(\eta)>0$ (iii). Then $l(\eta)=\eta\sum T(x_i)-nA(\eta)+\sum S(x_i)$ has $\ddot{l}=-n\ddot{A}<0$, and a root of $\dot{l}=\sum T(x_i)-n\dot{A}(\eta)$ is the MLE (Theorem 6.3). ■

In another parameter, $\eta=g(\theta)$ one-to-one, invariance gives: *match the mean of T to its sample average*, $E_\theta T(X_1)=\frac{1}{n}\sum T(x_i)$.

Example 6.7

(a) Bernoulli(p): $f=\exp\{x\log\frac{p}{1-p}+\log(1-p)\}$, $T(x)=x$; $\hat{p}=\bar{x}$ for $0<\bar{x}<1$.

(b) Poisson(λ): $T(x)=x$, $\eta=\log\lambda$; $\hat{\lambda}=\bar{x}$ when $\bar{x}>0$.

(c) Geometric $(1-p)^{x-1}p$, $x\geq 1$: $T(x)=x$, $\eta=\log(1-p)$; $E_pX=1/p=\bar{x}$ gives $\hat{p}=1/\bar{x}$ ($\bar{x}>1$).

(d) Exponential(θ): $T(x)=x$, $\eta=-1/\theta$; $\hat{\theta}=\bar{x}$.

(e) Pareto $\theta x^{-\theta-1}$, $x>1$: $T(x)=-\log x$; $\theta\log X\sim\mathrm{Exp}(1)$ (as $X^{-\theta}\sim U(0,1)$), so $E_\theta(-\log X)=-1/\theta$ and $\hat{\theta}=1/(\frac{1}{n}\sum\log x_i)$.

Without differentiability, study L directly; the MLE need not be unique.

Example 6.8 (Double exponential)

$f(x;\theta)=\frac{1}{2}e^{-|x-\theta|}$, $n=2m+1$: with $x_{(0)}=-\infty$, $x_{(n+1)}=\infty$, for $x_{(r)}\leq\theta<x_{(r+1)}$

$$l(\theta)=(n-2r)\theta+\sum_{i\leq r}x_{(i)}-\sum_{i>r}x_{(i)}-n\log 2.$$

The slope $n-2r$ is positive for $r\leq m$, negative for $r\geq m+1$: the MLE is the median $X_{(m+1)}$.

Example 6.9 (Uniform)

(a) $U(0,\theta)$: $L(\theta)=\theta^{-n}\mathbf{1}\{\theta\ge x_{(n)}\}$, so $\hat\theta^{\mathrm{MLE}}=X_{(n)}$ (no likelihood equation).
(b) $U(\theta-1,\theta+1)$: $L(\theta)=2^{-n}\mathbf{1}\{x_{(n)}-1\le\theta\le x_{(1)}+1\}$, so every statistic in $[X_{(n)}-1,X_{(1)}+1]$, e.g. $\alpha(X_{(n)}-1)+(1-\alpha)(X_{(1)}+1)$, is an MLE; other criteria must choose.

6.3 MLE for vector parameters

For $\theta\in\Omega\subseteq\mathbb{R}^k$ the definitions are the same; components of $\hat\theta^{\mathrm{MLE}}$ are MLEs of the components. Maximize one variable at a time, or solve $\partial l/\partial\theta_j=0$.

Example 6.10 (Double exponential with scale)

$f(x;\mu,\sigma)=\frac{1}{2\sigma}e^{-|x-\mu|/\sigma}$, $n=2m+1$:

$$l(\mu,\sigma)=-\frac{1}{\sigma}\sum|x_i-\mu|-n\log\sigma-n\log 2.$$

For each σ the median $\hat\mu=x_{(m+1)}$ maximizes in μ (Example 6.8), and

$$\frac{\partial}{\partial\sigma}l(\hat\mu,\sigma)=\frac{\sum|x_i-\hat\mu|}{\sigma^2}-\frac{n}{\sigma}>0\iff\sigma<\frac{1}{n}\sum|x_i-\hat\mu|,$$

so $\hat\sigma=\frac{1}{n}\sum|x_i-\hat\mu|$; the MLE is $(X_{(m+1)},\frac{1}{n}\sum|X_i-X_{(m+1)}|)$.

Example 6.11 (Normal)

$N(\mu,\sigma^2)$:

$$l(\mu,\sigma^2)=-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2-\frac{n}{2}\log\sigma^2-\frac{n}{2}\log 2\pi.$$

$\sum(x_i-\mu)^2=\sum(x_i-\bar x)^2+n(\bar x-\mu)^2$, so $\mu=\bar x$ for every σ^2; then $\partial_{\sigma^2}l(\bar x,\sigma^2)=\frac{1}{2\sigma^4}\sum(x_i-\bar x)^2-\frac{n}{2\sigma^2}>0$ iff $\sigma^2<\frac{1}{n}\sum(x_i-\bar x)^2$. Hence $\hat\mu=\bar X$, $\hat\sigma^2=\frac{1}{n}\sum(X_i-\bar X)^2$.

Theorem 6.7 (Invariance for functions)

If $\hat\theta^{\mathrm{MLE}}$ exists and $\eta_1=g_1(\theta)$ can be completed to a one-to-one map $\eta=(g_1(\theta),g_2(\theta))$, then $\hat\eta_1^{\mathrm{MLE}}=g_1(\hat\theta^{\mathrm{MLE}})$.

Proof. By Theorem 6.5 (verbatim for vectors), $\hat\eta^{\mathrm{MLE}}=(g_1(\hat\theta^{\mathrm{MLE}}),g_2(\hat\theta^{\mathrm{MLE}}))$; take the first component. ■

Example 6.12

(a) $N(\mu,\sigma^2)$: $(|\mu|,\operatorname{sgn}\mu,\sigma^2)$ is one-to-one, so $\widehat{|\mu|}^{\mathrm{MLE}}=|\bar X|$.
(b) Bernoulli: $(p(1-p),\operatorname{sgn}(p-\frac12))$ is one-to-one, so $\hat p(1-\hat p)$ is the MLE of $p(1-p)$.

The concavity results extend (same proofs):

Theorem 6.8

Let $\Omega_o = (a_1, b_1) \times \cdots \times (a_k, b_k)$ and let l have continuous second partial derivatives, with Hessian $\ddot{l}(\theta) = (\partial^2 l / \partial\theta_i \partial\theta_j)$.

(a) If $\mathbf{c}'\ddot{l}(\theta)\mathbf{c} < 0$ for all $\mathbf{c} \neq \mathbf{0}$ and $\theta \in \Omega_o$, and $\dot{l}(\hat{\theta}) = \mathbf{0}$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of l.

(b) If, moreover, $l(\theta) \to -\infty$ as θ tends to the boundary of Ω_o, then $\dot{l}(\theta) = \mathbf{0}$ has exactly one root in Ω_o, and it is the MLE.

Theorem 6.9 (Multiparameter exponential family)

Let $f(x;\eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$, $x \in \mathcal{X}$, $\eta \in N$, where (i) $\mathcal{X}$ does not depend on η;
(ii) N contains an open box in $\mathbb{R}^k$;
(iii) $\mathrm{Var}_\eta(\mathbf{c}'\mathbf{T}(X_1)) > 0$ for every $\mathbf{c} \neq \mathbf{0}$. If the likelihood equations

$$E_\eta T_j(X_1) = \frac{1}{n}\sum_{i=1}^n T_j(x_i), \qquad j = 1, \ldots, k,$$

have a root $\hat{\eta}$ in N, then $\hat{\eta}$ is the MLE.

Proof. As in Theorem 6.6, $E_\eta \mathbf{T} = \dot{A}(\eta)$, $\mathrm{Var}_\eta \mathbf{T} = \ddot{A}(\eta)$, so $\mathbf{c}'\ddot{l}(\eta)\mathbf{c} = -n\,\mathrm{Var}_\eta(\mathbf{c}'\mathbf{T}) < 0$; apply Theorem 6.8(a). ■

Example 6.13

(a) *Normal*: $\eta_1 = \mu/\sigma^2$, $\eta_2 = -1/(2\sigma^2)$, $T = (x, x^2)$; matching gives $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n}\sum x_i^2$, as in Example 6.11.
(b) *Bivariate normal*: $T = (x, y, xy, x^2, y^2)$; matching gives $\hat{\mu}_1 = \bar{x}$, $\hat{\mu}_2 = \bar{y}$, $\hat{\sigma}_j^2 = \frac{1}{n}\sum(\cdot - \text{mean})^2$, and $\hat{\rho}^{\text{MLE}}$ the sample correlation.
(c) *Multinomial* $\text{Multi}(n, \mathbf{p})$, one observation: with $\eta_j = \log(p_j/p_k)$, $j < k$, the pmf is $\exp\{\sum_{j<k}\eta_j x_j - nA(\eta) + S(x)\}$, $A(\eta) = \log(1 + e^{\eta_1} + \cdots + e^{\eta_{k-1}})$; $EX_j = np_j = x_j$ gives $\hat{p}_j = x_j/n$ (all $0 < x_j < n$).

6.4 Asymptotic properties of the MLE

Motivation. Under regularity the MLE is consistent and asymptotically normal with the Cramér–Rao variance.

Example 6.14

When the MLE is an MME, Chapter 5 applies:

- Bernoulli: $\hat{p} \xrightarrow{p} p$ and $\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1-p))$.
- Poisson: $\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N(0, \lambda)$. Exponential: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2)$.
- Normal: $\sqrt{n}(\hat{\mu} - \mu) \sim N(0, \sigma^2)$ exactly, $n\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-1)$, and $\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, 2\sigma^4)$.
- Bivariate normal: $\sqrt{n}(\hat{\rho} - \rho) \xrightarrow{d} N(0, (1-\rho^2)^2)$.

In general we argue directly. Let $\bar{l}_n(\theta) = \frac{1}{n}\sum_i \log f(X_i;\theta)$ and θ^0 the true value. By the LLN $\bar{l}_n(\theta) \xrightarrow{p} E_{\theta^0} \log f(X_1;\theta)$, so $\hat{\theta}_n = \arg\max \bar{l}_n$ should approach $\arg\max_\theta E_{\theta^0} \log f(X_1;\theta)$, which is θ^0:

Theorem 6.10 (Kullback–Leibler divergence)

Assume (R0) *identifiability*: $f(\cdot;\theta) = f(\cdot;\theta^0)$ implies $\theta = \theta^0$; and (R1) the support $\{x : f(x;\theta) > 0\} = \mathcal{X}$ does not depend on θ. Let

$$KL(\theta,\theta^0) = -E_{\theta^0} \log \frac{f(X;\theta)}{f(X;\theta^0)}.$$

Then (i) $KL(\theta,\theta^0) \geq 0$ for all θ, and (ii) $KL(\theta,\theta^0) = 0$ iff $\theta = \theta^0$. Equivalently, θ^0 is the unique maximizer of $E_{\theta^0} \log f(X;\theta)$.

Proof. $-\log t \geq 1 - t$ $(t > 0)$, with equality iff $t = 1$. With $t = f(X;\theta)/f(X;\theta^0)$,

$$KL(\theta,\theta^0) \geq E_{\theta^0}\Big[1 - \frac{f(X;\theta)}{f(X;\theta^0)}\Big] = 1 - \int_{\mathcal{X}} f(x;\theta)\,dx = 0,$$

by (R1); equality iff $f(\cdot;\theta) = f(\cdot;\theta^0)$ a.e., i.e. $\theta = \theta^0$ by (R0). ∎

A proof needs e.g. uniform convergence of $\bar{l}_n$ on compacts and continuity of $E_{\theta^0} \log f(X_1;\theta)$; these hold under strict concavity with $l_n \to -\infty$ at the boundary:

Theorem 6.11 (Consistency of the MLE)

Let Ω be an open box in $\mathbb{R}^k$ and assume (R0), (R1). Suppose l_n has continuous second partial derivatives with $\mathbf{c}'\ddot{l}_n(\theta)\mathbf{c} < 0$ for $\mathbf{c} \neq \mathbf{0}$ and $l_n(\theta) \to -\infty$ at the boundary of Ω, and that $E_{\theta^0} \log f(X_1;\theta)$ exists and is continuous in θ. Then the MLE $\hat{\theta}_n$ is the unique root of $\dot{l}_n(\theta) = \mathbf{0}$, and $\hat{\theta}_n \xrightarrow{p} \theta$ under P_θ for every $\theta \in \Omega$.

(Proof omitted.) E.g. the logistic MLE (Example 6.5) and the gamma MLE (root of $-\Psi(\alpha) - \log\beta + \overline{\log x} = 0$, $-\alpha/\beta + \bar{x}/\beta^2 = 0$, $\Psi = (\log\Gamma)'$) are consistent.

Heuristics for the limit law. Expanding the likelihood equation,

$$0 = \dot{\bar{l}}_n(\hat{\theta}_n) \approx \dot{\bar{l}}_n(\theta) + \ddot{\bar{l}}_n(\theta)(\hat{\theta}_n - \theta), \qquad \text{so} \qquad \sqrt{n}(\hat{\theta}_n - \theta) \approx \{-\ddot{\bar{l}}_n(\theta)\}^{-1}\sqrt{n}\,\dot{\bar{l}}_n(\theta).$$

and $-\ddot{\bar{l}}_n(\theta) \xrightarrow{p} I(\theta) := E_\theta[-\partial_\theta^2 \log f(X_1;\theta)]$ (LLN), while by the CLT

$$\sqrt{n}\,\dot{\bar{l}}_n(\theta) \xrightarrow{d} N\big(0, \mathrm{Var}_\theta\, \partial_\theta \log f(X_1;\theta)\big),$$

provided $E_\theta \partial_\theta \log f(X_1;\theta) = 0$.

Regularity conditions. In addition to (R0) and (R1):

(R2) Ω is an open subset of $\mathbb{R}^k$.

(R3) For every $x \in \mathcal{X}$, $\log f(x;\theta)$ has continuous first and second partial derivatives in θ.

(R4) For $u(\mathbf{X})$ with $E_\theta u(\mathbf{X})$ finite, $\partial_\theta^r E_\theta u(\mathbf{X}) = \int u(\mathbf{x}) \partial_\theta^r \prod f(x_i;\theta)\,d\mathbf{x}$, $r = 1, 2$.

(R5) The **Fisher information** $I(\theta) = \mathrm{Var}_\theta\,(\partial_\theta \log f(X_1;\theta))$ is finite and nonsingular.

$\dot{l}_n(\theta)$ is the **score function**.

Theorem 6.12 (Properties of the score)

Under (R0)–(R5),

(a) $E_\theta\{\partial_\theta \log f(X_1;\theta)\} = 0$;

(b) $I(\theta) = \mathrm{Var}_\theta\{\partial_\theta \log f(X_1;\theta)\} = E_\theta\{-\partial_\theta^2 \log f(X_1;\theta)\}$.

Proof. One-dimensional continuous θ: $\int_{\mathcal{X}} \exp\{\log f(x;\theta)\}\,dx = 1$ (R1); differentiating under the integral (R3, R4),

$$\int \{\partial_\theta \log f(x;\theta)\} f(x;\theta)\,dx = 0, \qquad \text{i.e.} \quad E_\theta \partial_\theta \log f(X_1;\theta) = 0.$$

and again,

$$\int \left\{\partial_\theta^2 \log f(x;\theta) + \left[\partial_\theta \log f(x;\theta)\right]^2\right\} f(x;\theta)\,dx = 0,$$

so $E_\theta[(\partial_\theta \log f)^2] = -E_\theta[\partial_\theta^2 \log f]$, the variance by (a). ∎

Theorem 6.13 (Asymptotic normality of the MLE)

Assume (R0)–(R5) and also:

(R6) the MLE $\hat\theta_n$ is the unique root of $\dot l_n(\theta) = 0$ and is consistent;

(R7) the third derivatives $\partial_\theta^3 \log f(x;\theta)$ exist and are continuous, and there is $M(x)$ with $\max_{\theta\in\Omega} \|\partial_\theta^3 \log f(x;\theta)\| \le M(x)$ and $E_\theta M(X_1) < \infty$.

Then $\sqrt{n}(\hat\theta_n - \theta) \xrightarrow{d} N(0, I(\theta)^{-1})$.

Proof. One-dimensional; $\dot{\bar l}_n = \dot l_n/n$, $\ddot{\bar l}_n = \ddot l_n/n$, $\bar l_n^{(3)} = l_n^{(3)}/n$.

Step 1 (expansion). By Taylor, with θ_n^* between θ and $\hat\theta_n$,

$$0 = \dot{\bar l}_n(\hat\theta_n) = \dot{\bar l}_n(\theta) + \ddot{\bar l}_n(\theta)(\hat\theta_n - \theta) + \tfrac{1}{2}\bar l_n^{(3)}(\theta_n^*)(\hat\theta_n - \theta)^2.$$

Step 2 (the remainder is negligible). Let $R_n = \frac{1}{2}\bar l_n^{(3)}(\theta_n^*)(\hat\theta_n - \theta)$. For $\epsilon, K > 0$,

$$P_\theta(|R_n| \ge \epsilon) \le P_\theta(|\bar l_n^{(3)}(\theta_n^*)| > K) + P_\theta(|\hat\theta_n - \theta| \ge 2\epsilon/K).$$

By (R7) and Markov, $P_\theta(|\bar l_n^{(3)}(\theta_n^*)| > K) \le E_\theta M(X_1)/K$; by (R6) the second term $\to 0$. So $\limsup_n P_\theta(|R_n| \ge \epsilon) \le E_\theta M(X_1)/K$ for all K: $R_n \xrightarrow{p} 0$.

Step 3 (linearization). $0 = \dot{\bar l}_n(\theta) + \{\ddot{\bar l}_n(\theta) + R_n\}(\hat\theta_n - \theta)$ and $-\ddot{\bar l}_n(\theta) \xrightarrow{p} I(\theta)$ (LLN, Theorem 6.12(b)), so

$$\sqrt{n}(\hat\theta_n - \theta) = \big(I(\theta) + r_n\big)^{-1} \sqrt{n}\,\dot{\bar l}_n(\theta), \qquad r_n \xrightarrow{p} 0.$$

Step 4 (limit). $\sqrt{n}\,\dot{\bar l}_n(\theta) \xrightarrow{d} N(0, I(\theta))$ (CLT, Theorem 6.12); by Slutsky, $\sqrt{n}(\hat\theta_n - \theta) \xrightarrow{d} I(\theta)^{-1} Z \sim N(0, I(\theta)^{-1})$. ∎

(R7) is stronger than necessary. Step 3 also gives

$$\sqrt{n}(\hat\theta_n - \theta) = I(\theta)^{-1} \sqrt{n}\,\dot{\bar l}_n(\theta) + o_p(1):$$

the MLE error is driven by the score.

One Newton–Raphson step from a good start is asymptotically as good:

Theorem 6.14 (One-step estimator)

Assume (R0)–(R7), and let $\hat\theta_n^{(0)}$ be an estimator with $\sqrt{n}(\hat\theta_n^{(0)}-\theta)\xrightarrow{d} Z$ for some random variable Z (R8). Then

$$\hat\theta_n^{(1)}=\hat\theta_n^{(0)}+\big[-\ddot l_n(\hat\theta_n^{(0)})\big]^{-1}\dot l_n(\hat\theta_n^{(0)})$$

satisfies $\sqrt{n}(\hat\theta_n^{(1)}-\theta)\xrightarrow{d}N(0,I(\theta)^{-1})$.

The proof follows Theorem 6.13, expanding $\dot l_n(\hat\theta_n^{(0)})$ around θ.

Example 6.15 (Logistic location)

Start from $\bar X$, with $\sqrt{n}(\bar X-\theta)\xrightarrow{d}N(0,\pi^2/3)$. The information is

$$I(\theta)=E_\theta\big[-\partial_\theta^2\log f(X;\theta)\big]=2E\Big[\frac{e^{-(X-\theta)}}{(1+e^{-(X-\theta)})^2}\Big]=2E[U(1-U)]=\frac13,$$

as $U=1/(1+e^{-(X-\theta)})\sim U(0,1)$. So $\sqrt{n}(\hat\theta^{(1)}-\theta)\xrightarrow{d}N(0,3)$, better than $\pi^2/3\approx3.29$.

Example 6.16 (Gamma)

Gamma(α,β), $\Psi=(\log\Gamma)'$:

$$\dot l_n=\begin{pmatrix}-n\Psi(\alpha)-n\log\beta+n\overline{\log x}\\ -n\alpha/\beta+n\bar x/\beta^2\end{pmatrix},\qquad I(\theta)=\begin{pmatrix}\dot\Psi(\alpha)&1/\beta\\1/\beta&\alpha/\beta^2\end{pmatrix}.$$

From the $\sqrt{n}$-consistent MME (Example 6.1(d), Theorem 6.2), one Newton step gives the limit $N_2(\mathbf{0},I(\theta)^{-1})$,

$$I(\theta)^{-1}=\frac{1}{\alpha\dot\Psi(\alpha)-1}\begin{pmatrix}\alpha&-\beta\\-\beta&\beta^2\dot\Psi(\alpha)\end{pmatrix}.$$

6.5 Least squares

In the **linear regression model**

$$Y_i=x_{i0}\beta_0+x_{i1}\beta_1+\cdots+x_{ip}\beta_p+e_i,\qquad Ee_i=0,\ \operatorname{Var}e_i=\sigma^2,\ \operatorname{Cov}(e_i,e_j)=0\ (i\neq j),$$

the **least squares estimator** (LSE) $\hat\beta$ minimizes $\sum_i\{Y_i-(x_{i0}\beta_0+\cdots+x_{ip}\beta_p)\}^2$. In matrix form $\mathbf{Y}=X\beta+\mathbf{e}$, X of size $n\times(p+1)$ and rank $p+1$, $E\mathbf{e}=\mathbf{0}$, $\operatorname{Var}\mathbf{e}=\sigma^2I_n$, and

$$\hat\beta^{\mathrm{LSE}}=\operatorname*{arg\,min}_{\beta\in\mathbb{R}^{p+1}}\|\mathbf{Y}-X\beta\|^2.$$

Theorem 6.15 (Least squares)

(a) $\Pi=X(X'X)^{-1}X'$ satisfies $\Pi'=\Pi$, $\Pi'(I-\Pi)=0$ and $\Pi X=X$.

(b) $\|\mathbf{Y}-X\beta\|^2=\|\Pi(\mathbf{Y}-X\beta)\|^2+\|(I-\Pi)\mathbf{Y}\|^2$; hence $X\hat\beta^{\mathrm{LSE}}=\Pi\mathbf{Y}$ and $\hat\beta^{\mathrm{LSE}}=(X'X)^{-1}X'\mathbf{Y}$.

Proof. (a) $\Pi'=X\{(X'X)^{-1}\}'X'=\Pi$ and

$$\Pi'\Pi=X(X'X)^{-1}X'X(X'X)^{-1}X'=\Pi,$$

so $\Pi'(I-\Pi)=0$, and $\Pi X = X$.

(b) $\mathbf{Y} - X\beta = \Pi(\mathbf{Y}-X\beta) + (I-\Pi)\mathbf{Y}$ with vanishing cross term; the second part is free of β, and the first is zero iff $X\beta = \Pi\mathbf{Y}$, i.e. $\beta = (X'X)^{-1}X'\mathbf{Y}$. ■

Π projects orthogonally onto $\mathrm{col}(X)$. The error variance is estimated by

$$\hat\sigma^2 = \frac{\|\mathbf{Y} - X\hat\beta\|^2}{n-p-1}.$$

Theorem 6.16 (Properties of the LSE)

(a) $E\hat\beta = \beta$ and $\mathrm{Var}\,\hat\beta = \sigma^2(X'X)^{-1}$.

(b) $E\hat\sigma^2 = \sigma^2$.

(c) If $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$, then $\hat\beta \sim N(\beta, \sigma^2(X'X)^{-1})$, $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and they are independent.

Proof. (a) $E\hat\beta = (X'X)^{-1}X'X\beta = \beta$ and $\mathrm{Var}\,\hat\beta = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $(I-\Pi)\mathbf{Y} = (I-\Pi)\mathbf{e}$, so $(n-p-1)\hat\sigma^2 = \mathbf{e}'(I-\Pi)\mathbf{e}$ and

$$E\,\mathbf{e}'(I-\Pi)\mathbf{e} = E\mathrm{tr}\{(I-\Pi)\mathbf{e}\mathbf{e}'\} = \mathrm{tr}\{(I-\Pi)\sigma^2 I\} = \sigma^2\{n - \mathrm{tr}((X'X)^{-1}X'X)\} = \sigma^2(n-p-1).$$

(c) is Theorem 4.19 in Volume I. ■

Orthogonalization. Partition $X = (X_0, X_1)$ (p_0 and p_1 columns), $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$. Then $X_{1|0} = (I-\Pi_0)X_1$ is orthogonal to X_0, and

$$X\beta = X_0\beta_0 + X_1\beta_1 = X_0\gamma_0 + X_{1|0}\beta_1, \qquad \gamma_0 = \beta_0 + (X_0'X_0)^{-1}X_0'X_1\beta_1.$$

With $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$, $\Pi = \Pi_0 + \Pi_{1|0}$ and $\Pi_0\Pi_{1|0} = 0$.

Theorem 6.17

In this notation,

(a) $X_0\hat\gamma_0 = \Pi_0\mathbf{Y}$, $X_{1|0}\hat\beta_1 = \Pi_{1|0}\mathbf{Y}$, i.e. $\hat\gamma_0 = (X_0'X_0)^{-1}X_0'\mathbf{Y}$ and $\hat\beta_1 = (X_{1|0}'X_{1|0})^{-1}X_{1|0}'\mathbf{Y}$, and $\hat\beta_0 = \hat\gamma_0 - (X_0'X_0)^{-1}X_0'X_1\hat\beta_1$;

(b) $\mathrm{Cov}(\hat\gamma_0, \hat\beta_1) = 0$, $\mathrm{Var}\,\hat\gamma_0 = \sigma^2(X_0'X_0)^{-1}$ and $\mathrm{Var}\,\hat\beta_1 = \sigma^2(X_{1|0}'X_{1|0})^{-1}$.

Proof. (a) The design $(X_0, X_{1|0})$ has block-diagonal Gram matrix, so $(X'X)^{-1}X'\mathbf{Y}$ splits; invert the definition of γ_0 for $\hat\beta_0$.

(b) $\mathrm{Cov}(\hat\gamma_0, \hat\beta_1) = \sigma^2(X_0'X_0)^{-1}X_0'X_{1|0}(X_{1|0}'X_{1|0})^{-1} = 0$; variances by Theorem 6.16(a). ■

Example 6.17 (Centring the explanatory variables)

$EY_i = \gamma_0 + \sum_j \beta_j(x_{ij} - \bar x_{\cdot j})$, $\gamma_0 = \beta_0 + \sum_j \beta_j \bar x_{\cdot j}$: $X_0 = \mathbf{1}$, $X_{1|0}$ has entries $x_{ij} - \bar x_{\cdot j}$. With $S_{jk} = \sum_i (x_{ij} - \bar x_{\cdot j})(x_{ik} - \bar x_{\cdot k})$, $S_{jY} = \sum_i (x_{ij} - \bar x_{\cdot j})(Y_i - \bar Y)$,

$$\hat\gamma_0 = \bar Y, \qquad \begin{pmatrix} S_{11} & \cdots & S_{1p} \\ \vdots & & \vdots \\ S_{p1} & \cdots & S_{pp} \end{pmatrix} \begin{pmatrix} \hat\beta_1 \\ \vdots \\ \hat\beta_p \end{pmatrix} = \begin{pmatrix} S_{1Y} \\ \vdots \\ S_{pY} \end{pmatrix}.$$

For $p = 1$: $\hat{\beta}_1 = S_{1Y}/S_{11}$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, $\operatorname{Var} \hat{\beta}_1 = \sigma^2/S_{11}$, $\operatorname{Cov}(\bar{Y}, \hat{\beta}_1) = 0$.